

Scaling Book Exercises: Section 1

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Source scan: ../handwritten/section_01_handwritten.pdf

Exercise 1

Consider the matrix multiplication

$$X \in \mathbb{R}^{B \times D}, \quad Y \in \mathbb{R}^{D \times F}, \quad Z = XY \in \mathbb{R}^{B \times F}.$$

- (a) Loading the inputs costs

$$BD + DF$$

bytes, and writing the output costs

$$BF$$

bytes.

- (b) The total number of floating-point operations is

$$2BDF.$$

- (c) The arithmetic intensity is therefore

$$\text{AI} = \frac{2BDF}{BD + DF + BF}.$$

- (d) Using a tensor-core throughput of 3.94×10^{14} FLOPs/s and communication bandwidth of 8.2×10^{11} bytes/s,

$$T_{\text{math}} = \frac{2BDF}{3.94 \times 10^{14}}, \quad T_{\text{comm}} = \frac{BD + DF + BF}{8.2 \times 10^{11}}.$$

A reasonable upper bound on runtime is

$$T_{\text{math}} + T_{\text{comm}},$$

and a reasonable lower bound is

$$\max(T_{\text{math}}, T_{\text{comm}}).$$

Exercise 2

We solve for the batch size at which compute time and communication time are equal. Assume

$$B \ll D, \quad B \ll F.$$

Using the arithmetic intensity threshold from the notes,

$$\frac{2BDF}{DF} = 2B > 240.$$

Therefore,

$$B > 120.$$

Exercise 3

For a fixed batch size $B = 100$, the total bytes moved are approximately

$$2BD + 2BF + DF.$$

When $D = F$, this becomes

$$4BD + D^2.$$

The arithmetic intensity is

$$\text{AI} \approx \frac{2BD^2}{2BD + 2BF + DF} = \frac{2BD^2}{4BD + D^2} = \frac{2BD}{4B + D}.$$

With $B = 100$,

$$\text{AI} = \frac{200D}{400 + D}.$$

For $D = F = 4096$,

$$\text{AI} = \frac{200 \cdot 4096}{400 + 4096} \approx 182.$$

For $D = F = 1024$,

$$\text{AI} = \frac{200 \cdot 1024}{400 + 1024} \approx 143.$$

Exercise 4

Arithmetic intensity is

$$\text{AI} = \frac{\text{FLOPs}}{\text{communication}}.$$

For the matrix multiplication

$$[B, D] @ [D, F],$$

the FLOP count is

$$2BDF.$$

The byte count is

$$BD + BDF + BF.$$

Thus

$$\text{AI} = \frac{2BDF}{BD + BDF + BF}.$$

Assuming $B \ll D$ and $B \ll F$, the dominant term in the denominator is BDF , so

$$\text{AI} \approx \frac{2BDF}{BDF} = 2.$$

This is constant with respect to the matrix sizes under these assumptions.

Exercise 5

The critical batch size is computed from the tensor-core throughput and bandwidth:

$$\text{tensor core throughput} = 1979 \text{ teraFLOPs}, \quad \text{bandwidth} = 3.35 \text{ terabytes/s}.$$

Accounting for multiply-add as two FLOPs,

$$B_{\text{critical}} = \frac{1979/2}{3.35} \approx 295.37.$$

Thus,

$$B_{\text{critical}} \approx 295.$$

Verification Notes

The formulas above are converted from the handwritten scan. Before final submission, verify the following against the exercise text:

- Exercise 4's byte-count expression, since the handwritten denominator appears as $BD + BDF + BF$.
- Whether the bandwidth constants should be expressed in bytes/s or words/s in the final writeup.